

# BCSE204L Design and Analysis of Algorithms

Module-2

Topic-2: Longest Common Subsequence  
(LCS)

# Longest Common Subsequence (LCS)

- The longest common subsequence (LCS) problem is the problem of finding the longest subsequence common to all sequences in a set of sequences
- Unlike substrings, subsequences are **NOT** required to occupy consecutive positions within the original sequences

# LCS Example

- Seq1: ( $ABCD$ ), Seq2: ( $ACBAD$ )
- Length-2 common subsequences:  
( $AB$ ), ( $AC$ ), ( $AD$ ), ( $BD$ ), and ( $CD$ ), totaling to 5
- Length-3 common subsequences: ( $ABD$ ) and ( $ACD$ ), totaling to 2
- No common subsequences longer than 3
- So ( $ABD$ ) and ( $ACD$ ) are their longest common subsequences and length of LCS = 3

# LCS Demo

- We will do the demo first to understand the concept/theory better
- Demo on board

## LCS DP- The idea

- Let the input sequences be  $X[0..m - 1]$  and  $Y[0..n - 1]$  of lengths  $m$  and  $n$  respectively.
- Let  $L(X[0..m - 1], Y[0..n - 1])$  be the length of LCS of the two sequences  $X$  and  $Y$ .
- The recursive definition of  $L(X[0..m - 1], Y[0..n - 1])$  can be framed based on whether the last characters in the previous call match or not

# LSP DP- The idea

- If last characters of both sequences match (or  $X[m - 1] == Y[n - 1]$ ) then
$$\begin{aligned}L(X[0..m - 1], Y[0..n - 1]) \\&= 1 + L(X[0..m - 2], Y[0..n - 2])\end{aligned}$$
- If last characters of both sequences do not match (or  $X[m - 1] \neq Y[n - 1]$ ) then
$$\begin{aligned}L(X[0..m - 1], Y[0..n - 1]) = \\&\text{MAX}\{L(X[0..m - 2], Y[0..n - 1]), L(X[0..m \\&- 1], Y[0..n - 2])\}\end{aligned}$$

# LCS DP- An Example

- Consider the input strings “*AGGTAB*” and “*GXTXAYB*”.  
Last characters match for the strings

- So length of LCS can be written as:

$$L(AGGTAB, GXTXAYB) = 1 + L(AGGTA, GXTXAY)$$

- Consider the input strings “*ABCDGH*” and “*AEDFHR*”.  
Last characters do not match for the strings

- So length of LCS can be written as:  $L(ABCDGH, AEDFHR) = \text{MAX} \{L(ABCDG, AEDFHR), L(ABCDGH, AEDFH)\}$

# LCS DP Pseudocode

```
X and Y be two given sequences
Initialize a table LCS of dimension X.length * Y.length
X.label = X
Y.label = Y
LCS[0][] = 0
LCS[][0] = 0
Start from LCS[1][1]
Compare X[i] and Y[j]
    If X[i] = Y[j]
        LCS[i][j] = 1 + LCS[i-1, j-1]
    Else
        LCS[i][j] = max(LCS[i-1][j], LCS[i][j-1])
```



# Analysis

- Time complexity
- If  $m$  and  $n$  are the lengths of the sequences
- Then filling an  $m \times n$  table
- Therefore,  $O(mn)$

Thank you..